MZUZU UNIVERSITY | FACULTY OF SCIENCE, TECHNOLOGY & INNOVATION

DEPARTMENT OF INFORMATION & COMMUNICATION TECHNOLOGY (ICT) - SOFTWARE EXHIBITION 2026

FARMPULSE MALAWI

Data-Driven Climate Intelligence, Crop Planning & Parametric Risk Protection for Malawi's Sma
Extension Systems

THE BIG IDEA

Transforming smallholder agriculture from high-risk climate guesswork into predictable, profitable, and de-risked a
localized agromet intelligence, satellite telemetry, and governance tools.

THE PROBLEM

THE REAL-WORLD CHALLENGE

Malawi's 3.2 million smallholder farming households invest scarce capital into seed and fertilizer with ZERO reliable data on rainfall continuation, crop economic viability, or market commodity prices.

Farming blind results in catastrophic seasonal failure across rural communities.

HOW IT IS HANDLED TODAY

Farmers rely entirely on intuition, guesswork, and delayed radio weather broadcasts.

Government extension officers face impossible coverage ratios: 1 field officer to over 2,500 farmers.

Personalized, timely agronomic guidance is physically impossible to deliver in person.

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A single 14-da
60% of

Agriculture ac
earning

Yet 74% of sm
to forward

OUR SOLUTION

INTEGRATED AGRICULTURAL INTELLIGENCE

FarmPulse Malawi is an integrated agricultural decision-support and governance ecosystem that unites:

- Localized agro-ecological climate risk modeling
- Dynamic enterprise crop and budget planning
- Sentinel-2 Earth Observation satellite telemetry
- Zero-adjuster parametric micro-crop insurance

Transforming raw data into clear, profit-maximizing planting strategies.

WHAT USERS CAN

BETTER
Compare Food Security vs. High-Mar
purchasing seed c

FASTER
Generate full seasonal input allocations
prices in under 90 secon

MORE AFFOR
De-risk farm investments with automat
pays out straight to Airtel Mo

WHO BENEFITS?

PRIMARY USERS

Smallholder Farmers (Web & *384# USSD):
Commercial and semi-commercial farmers accessing real-time weather alerts, crop budgeting, and market pricing in English, Chichewa, and Tumbuka.

District Agricultural Officers (DADOs):
Ministry extension supervisors monitoring district-wide crop health, dry spells, and dispatching SMS broadcasts.

Farmer Cooperatives & Outgrower Unions:
Managing collective input buying and grain aggregation.

BENEFICIARIES & WID

Agribusiness Off-Takers (Py
Gain supply-chain traceability and accur
weeks before co

Lending Banks & MFIs (N
De-risk seasonal loans using verified 5-
default rates from 16%

Parametric Insurers (NICO G
Sell automated drought insurance

HOW IT WORKS

STEP 1

USER INPUT

Farmer enters land size (ha/ac), seasonal budget (MWK), district/AEZ, soil type, and preferred crops via Web or *384# USSD.

STEP 2

INTELLIGENCE ENGINE

Engine cross-references 10-year district rainfall records, dry-spell risk curves, input costs (AIP rates), and regional commodity market spreads.

STEP 3

PRODUCT OUTPUT

Delivers 3 tailored allocations (Food Security, High Margin, Climate Resilient) with input quantities, planting dates, and breakeven market prices.

PRODUCT SNAPSHOT

SMALLHOLDER INTELLIGENCE

Interactive crop planner and farm history console.

- 10-year longitudinal yields
- Dynamic land allocator
- Breakeven price solver
- Fertilizer subsidy calculator

Route: /planner/ & /farm-history/

SENTINEL-2 DIGITAL TWIN

Multi-spectral Earth Observation satellite tracking.

- NDVI canopy vigor (0.0 to 1.0)
- Topsoil moisture index (0-5cm)
- Cadastral parcel polygons
- Automated stress alerts

Route: /satellite/

DADO

National o

- 15 f
- 48.5
- Clim
- Emer

Route:

STANDOUT FEATURES

DYNAMIC LAND & BUDGET ALLOCATOR
-> Benefit: Balances seed and fertilizer costs against capital, guaranteeing farmers never run out of money mid-season.

SENTINEL-2 SATELLITE DIGITAL TWIN
-> Benefit: Detects moisture stress and canopy deficit from space every 5 days without purchasing expensive field IoT sensors.

ZERO-ADJUSTER PARAMETRIC INSURANCE
-> Benefit: Uses automated satellite rainfall indexing to dispatch instant mobile money claims with zero corrupt adjusters.

BASIC PHONE ACCESSIBIL
-> Benefit: Delivers agromet forecast
Chichewa & Tumbuka to farm

DADO EXTENSION & GOV
-> Benefit: Equips Ministry leaders with
aggregated yield projections, and e

MARKET TRANSPORT AR
-> Benefit: Compares prices across 6 t
Limbe), proving if transp

PROOF IT WORKS

15
ACTIVE FARMERS
Across 12 Districts

48.5 ha
MONITORED LAND
100% Satellite Mapped

194
PLANTING RECORDS
10-Year Longitudinal History

CURRENT DEVELOPMENT STAGE: LIVE MVP / ACTIVE PRE-PILOT

EVIDENCE & FIELD VALIDATION:
- Validated across all 3 agricultural regions of Malawi (Mzuzu Highlands, Central Plain, Shire Vall
- 10 full seasons of historical performance (2015/16 through 2024/25) verified on Miriam Chiphaka Fie

WHAT MAKES IT DIFFERENT?

TRADITIONAL AG APPS

- Broadcast generic weather forecasts or static price tables.

- Tell the farmer it will rain 10mm, leaving them to guess what that means for fertilizer application or crop survival.

- Disconnected point solutions that ignore the financing and insurance barrier.

THE FARMPULSE EDGE

- Translates data into financial consequence: tells the farmer exactly what crops to plant, how to budget inputs, and when to sell.

- Calibrates 10-year district climate risk against individual land size and cash constraints.

- Automated parametric insurance integrated into the planting plan.

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IMPACT & VALUE

< 90 SECONDS
TIME SAVED

Reduces multi-crop agronomic budgeting and scenario planning from 3 weeks of manual field visits down to under 90 seconds of algorithmic generation.

Enables 1 extension officer to support 2,500+ farmers digitally.

22% CUT
INPUT COST REDUCED

Eliminates input expenditure waste by 22% through precise seed and basal/top-dressing fertilizer ratios.

Slashes bank agricultural loan default rates from 16.5% to under 4% through verified yield histories.

Boosts sm
margins by
legume ro
region

Lifts families

SUSTAINABILITY & BUSINESS MODEL

HYBRID B2B2C & B2B2G MONETIZATION

1. Agribusiness Outgrower SaaS:
Off-takers (Pyxus, ETG, Sunseed) pay MWK 4.5M - 12M/yr for smallholder tracking & yield forecasting.

2. Fintech Loan Risk-Scoring API:
Commercial banks & MFIs (NBS Bank, FINCA) pay 1.75% fee per de-risked seasonal input loan.

3. Insurtech Brokerage Commission:
8% to 10% technology fee on gross written insurance premiums underwritten by NICO General / Old Mutual.

4. Government DPI Licenses:
Ministry of Ag & WFP district analytics (MWK 15M - 25M/district).

3-YEAR FINANCIAL PR

YEAR 1 (Pilo
- Active Smallholder
- Gross Revenue: MWK 42
- Operating Costs: M
- EBITDA: +MWK 11,500
-> BREAKEVEN ACHIEV

YEAR 2 (Region
- Active Smallholders
- Gross Revenue: MWK 17
- Operating Costs: M
- EBITDA: +MWK 85,200

NEXT MILESTONES

NEXT 3 MONTHS
Q1 2027

FIELD PILOT VALIDATION

- Formalize pilot integration with 3 farmer cooperatives in Mzuzu and Lilongwe (500 smallholders).

- Deploy official commercial short-code for USSD *384# with TNM and Airtel.

- Publish localized crop budget matrices for Mzuzu Highlands AEZ.

NEXT 6 MONTHS
Q2 - Q3 2027

FINTECH & INSURANCE SANDBOX

- Complete API sandbox testing with NBS Bank for agricultural credit scoring.

- Partner with NICO General for commercial parametric micro-insurance underwriting.

- Integrate Mlimi AI Chichewa voice-to-text advisory prototype.

- Scale p
smallholde

- Secure inst
Ministry of

- Expand s
Sentinel-1 S

WHAT WE NEED FROM YOU

WE ARE ACTIVELY LOOKING FOR:

[X] PILOT SITES & COOPERATIVES
3 commercial agricultural cooperatives or outgrower schemes for structured Phase 1 field testing.

[X] INSTITUTIONAL PARTNERS
Underwriting insurers (NICO General, Old Mutual) and agricultural lenders (NBS Bank, FINCA).

[X] FACULTY MENTORS
Mzuzu University agronomy and economics lecturers to validate regional agro-ecological zone yield models.

[X] SEED GRANTS & TECHNICAL SUPPORT

OUR ONE CLE

"Partner with FarmPulse today to
resilience, de-risked finance, and
smallholder farming commu

LIVE DEMO PLA
https://farmpulse-p6j

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